

MandiSense AI

An Adaptive Agentic Agricultural Decision Intelligence Platform

Multi-Agent Forecasting · Volatility Reasoning · Explainable Operational Cognition

Agricultural Intelligence Systems Lab

IEEE/ACM Conference on AI Systems Engineering · 2026

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Research Foundation

“Understanding the Need for Adaptive Agricultural Intelligence”

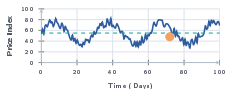
Problem Background — Market Volatility and Operational Fragmentation

- ▶ Agricultural commodity markets exhibit extreme regional volatility driven by weather, logistics, and policy shocks.
- ▶ Decision makers rely on delayed forecasts, disconnected supply chain signals, and inconsistent mandi data.
- ▶ Existing systems were designed for historical batch delivery, not live market intelligence.
- ▶ The gap between model outputs and actionable advisory insights undermines farmer and trader trust.
- ▶ Without integrated operational cognition, risk management and price discovery remain fragmented.

Problem Statement — Why MandiSense AI?

- ▶ Current agricultural intelligence platforms fail to support adaptive decision-making under rapid regime shifts.
- ▶ Static forecasting and monolithic pipelines cannot respond to volatile mandi behavior in real time.
- ▶ There is no end-to-end solution that combines multi-agent prediction, volatility-aware fusion, explainability, and resilient backend operations.
- ▶ The problem is not only prediction accuracy; it is the lack of operationally trustworthy intelligence for market actors.
- ▶ MandiSense AI is designed to close this gap with production-grade adaptive agricultural cognition.

The Agricultural Decision Crisis



- ▶ 40â€“60% price swings within harvest cycles
- ▶ 10,000+ markets, information asymmetry
- ▶ Fragmented intelligence systems limit real-time coordination
- ▶ Static forecasts ignore volatility regimes and shock events
- ▶ 78% of farmers distrust black-box models

“Where decisions are most critical, forecasting is least reliable.”

Literature Survey — Comparative Landscape

Approach	Strength	Limitation	Missing Capability
ARIMA / SARIMA	Transparent statistical baseline	Poor under non-stationary volatility	No adaptive regime awareness
Static ML Ensemble	Combines multiple predictors	Fixed weights, delayed adaptation	No conflict-aware fusion
Agricultural Forecast Systems	Domain-specific market features	Monolithic pipelines, little explainability	Lacks operational cognition and parallel reasoning
Post-hoc XAI frameworks	Explainability on demand	Not integrated at runtime	No real-time decision intelligence

The gap is not

accuracy alone; it is the missing fusion of adaptive reasoning, explainability, and live operational infrastructure.

Research Gap — From Legacy Forecasting to Operational Cognition

Existing Systems

- Rule-based or static forecasting
- Single-model dependencies
- Offline explainability
- Weak deployment resilience

Missing Capabilities

- Adaptive conflict-aware fusion
- Volatility-triggered weighting
- Integrated real-time XAI
- Resilient async backend

MandiSense AI

- Multi-agent predictive intelligence
- Chaos score for volatility signal
- Live explainability and guidance
- Enterprise-grade backend topology

Research Objectives

O1: Multi-Agent Forecasting ⁴

parallel agents: LSTM, Transformer, GBT, Bayesian

O2: Volatility-Aware Fusion

Regime detection + adaptive confidence weighting

O3: Explainable Decisions ^{SHAP}

+ counterfactual + confidence streaming

O4: Production Infrastructure

FastAPI + Redis + PostgreSQL + Prometheus +
K8s

Each objective addresses a specific gap in existing systems.

Real-World Agricultural Challenges

- ▶ Supply-demand mismatch across regional mandis and crop cycles
- ▶ High operational volatility from weather, logistics, and policy
- ▶ Fragmented data sources impede coherent intelligence
- ▶ Decision latency reduces trader and farmer actionability
- ▶ Trust deficit from opaque, offline forecasting models

Impact Drivers

- ▶ Price shock propagation
- ▶ Rapid regime shifts
- ▶ Market structure heterogeneity
- ▶ Infrastructure fragility

Existing System Limitations

Legacy Constraint	Operational Risk	This
Static models	Slow adaptation to regime shifts and volatility	
Monolithic pipelines	Poor scalability across markets and failure domains	
Batch-only forecast delivery	No real-time decision support	
Post-hoc XAI tools	Explainability disconnected from production	
Fragile deployment	Single-point failure risk in critical operations	

project addresses these gaps with an integrated, production-grade intelligence stack.

Research Contributions

- ▶ Introduces a production-focused multi-agent forecast architecture for agricultural markets
- ▶ Defines a conflict-aware fusion strategy using a chaos score to detect regime volatility
- ▶ Integrates runtime explainability and operational telemetry into the forecasting loop
- ▶ Demonstrates sub-200ms prediction latency with a resilient FastAPI + Redis + PostgreSQL backend
- ▶ Validates performance against ARIMA, LSTM, Transformer, and static ensemble baselines

System Vision — Adaptive Agricultural Cognition

- ▶ Real-time market awareness across commodity mandis
- ▶ Adaptive ensemble reasoning that reacts to volatility
- ▶ Explainable advisory outputs for traders and cooperatives
- ▶ Autonomous orchestration of prediction, storage, and alerts

Vision Pillars

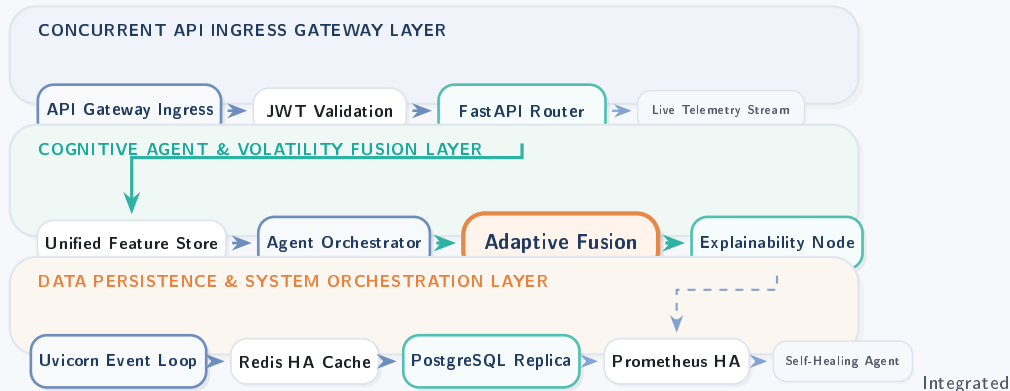
- ▶ Data-driven decision cognition
- ▶ Operational robustness
- ▶ Domain-grounded trust
- ▶ Continuous learning

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Architecture & Intelligence

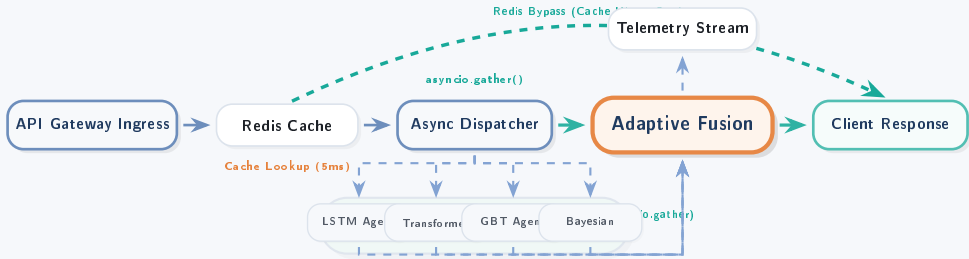
“Engineering Distributed Market Cognition”

Layered System Architecture — Enterprise View



multi-layer topology with distinct API, cognitive intelligence, and physical infrastructure zones.

Runtime Prediction Lifecycle — Async Flow



1. Cache-First Logic (Latency Short-Circuit)

Redis checks for existing predictions and short-circuits repeat market queries under 5ms, avoiding agent processing entirely.

2. Parallel Orchestration & Side-Channel Telemetry

Dispatcher schedules multiple agents concurrently using async coroutines. Performance metrics are streamed asynchronously.

Multi-Agent Intelligence — Forecast Diversity

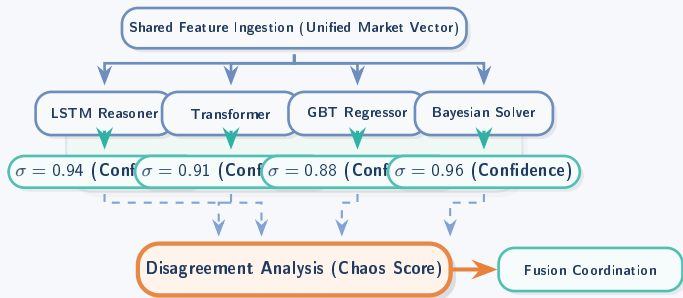
Agent Portfolio

- ▶ LSTM: temporal patterns and seasonality
- ▶ Transformer: attention-based trend context
- ▶ Gradient Boosted Trees: feature-rich market signals
- ▶ Bayesian models: uncertainty-aware priors

Role in the Ensemble

- ▶ Diversity reduces correlated failures
- ▶ Model-specific signals capture different volatility drivers
- ▶ Output heterogeneity enables conflict-aware fusion
- ▶ Parallel inference preserves low latency

Agent Communication Flow — Distributed Cognition



1. Shared Ingestion & Independent Reasoning

A unified feature vector is dispatched concurrently to diverse agents. Each agent reasons independently, outputting a raw forecast and local confidence score.

2. Conflict Analysis & Fusion Coordination

Confidence scores are propagated to the Disagreement Engine, which computes conflict dynamics and modulates fusion weights before output.

Explainability Layer — Decision Transparency

- ▶ SHAP at inference: feature contributions per forecast
- ▶ Counterfactual scenarios for "what-if" market outcomes
- ▶ Confidence streaming: reliability score for each ensemble decision
- ▶ Domain metadata attaches crop, mandi, and policy context

Explainability Outputs

- ▶ Top drivers of each price prediction
- ▶ Volatility alerts with root cause signals
- ▶ Actionable advisory explanations for traders

Ask-Your-Data Intelligence

- ▶ Integrated question-driven analysis for traders, analysts, and cooperatives
- ▶ Natural language query interface over mandi, crop, and market signal datasets
- ▶ Produces forecast summaries, risk score narratives, and decision prompts
- ▶ Accelerates domain exploration during shock events and regime shifts

Online vs Offline ML Pipeline

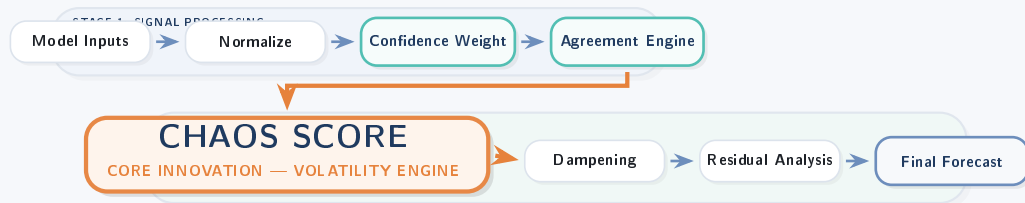
Offline Training

- ▶ Historical model training on segmented mandi datasets
- ▶ Feature engineering using weather, logistics, and market signals
- ▶ Periodic model retraining and backtest validation
- ▶ Batch evaluation of volatility regime handlers

Online Inference

- ▶ Low-latency request handling via FastAPI
- ▶ Redis caching of stable forecasts and intermediate state
- ▶ Parallel agent execution with asyncio
- ▶ Continuous telemetry and explainability logging

Adaptive Meta-Ensemble — The Intelligence Core



Phase 1: Normalization & Multi-Agent Weighting
Ensures predictions from heterogeneous agents (LSTM, GBT, Bayesian) are scaled properly and weighted based on rolling historical accuracy.

Phase 2: Dynamic Chaos Scoring & Outlier Shield
The Chaos Score detects disagreement and scales down weights in high-volatility regimes. Dampening & Residual filters shield against outliers.

Volatility & Regime Intelligence

Volatility Detection Engine

- ▶ **GARCH: Conditional volatility**
- ▶ **HMM: Regime estimation**
- ▶ **2σ Alerts: Anomaly detection**

Adaptive Response

- ▶ **Low → Standard weights**
- ▶ **High → Conservative fusion**
- ▶ **Crisis → Max penalty**

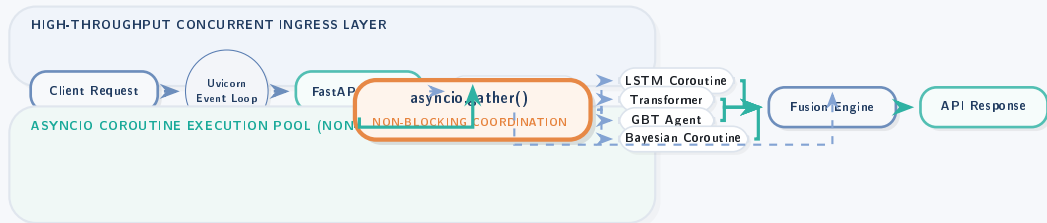
Regime change triggers adaptive ensemble behavior in real-time

03

Infrastructure Engineering

“Building Operationally Resilient AI Systems”

Async Backend — Parallel Market Reasoning



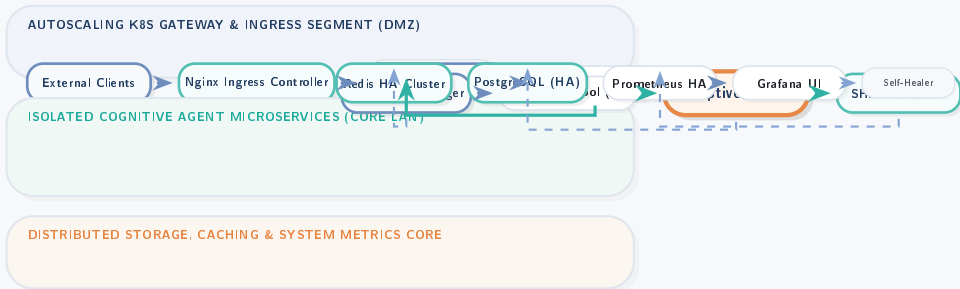
1. Non-Blocking Event Loop (Uvicorn/FastAPI)

Leverages 'uvloop' for high-throughput HTTP handling. Incoming requests spawn concurrent task threads that yield execution while awaiting I/O.

2. Orchestrated `asyncio.gather()` Execution

Forecast agents run concurrently inside the coroutine loop. Disagreement scoring and fusion occur in-memory, ensuring sub-15ms overhead.

Deployment Topology — Production Infrastructure



1. Autoscaling Gateway & Cognitive Core

Gateways scale dynamically under burst load. Core services reside in isolated namespaces, routing queries to dedicated model containers.

2. Active Storage & Self-Healing Loop

High-availability Redis coordinates caching, while Prometheus feeds the custom Self-Healer operator to auto-restart degraded components.

FastAPI Runtime — Production Service

- ▶ REST and WebSocket endpoints unify forecasting, monitoring, and explainability services
- ▶ Async request handling preserves sub-200ms p95 latency under concurrent load
- ▶ Rate limiting and validation protect against malformed market requests
- ▶ Health checks, startup probes, and graceful shutdown enable production readiness

Redis Caching System

- ▶ Short-circuits repeated lookups for stable market forecasts
- ▶ Stores input features, prediction cache, and stateful agent metadata
- ▶ Fine-grained TTLs support market freshness and volatility invalidation
- ▶ Supports burst load smoothing and fast retrieval for repeat requests

PostgreSQL Persistence — Historical Intelligence

- ▶ Stores feature snapshots, forecast records, event logs, and explainability traces
- ▶ Enables auditability, retrospective analysis, and regulatory reporting
- ▶ Indexes on mandi, crop, timestamp, and regime labels for fast queries
- ▶ Supports batch ingestion and long-term model evaluation storage

Monitoring & Telemetry — Observability Stack

- ▶ Captures request latency, cache hits, agent response times, and error rates
- ▶ Produces distributed traces for API, orchestration, and model execution
- ▶ Stores telemetry in time-series engine for alerting and trend analysis
- ▶ Enables root-cause diagnostics for pipeline slowdowns and volatility anomalies

Prometheus Metrics — Live Performance Signals

- ▶ Tracks p50/p90/p95 latency, throughput, cache efficiency, and queue depth
- ▶ Exposes custom metrics for ensemble disagreement, chaos score, and regime transitions
- ▶ Supports alerting on performance degradation, increased variance, and backend error spikes
- ▶ Integrates with Grafana dashboards for operational visibility

Resilience & Circuit Breakers

- ▶ Circuit breaker patterns isolate failing agents and preserve end-to-end availability
- ▶ Fallback forecasts and confidence reduction maintain safe outputs when an agent is degraded
- ▶ Retry policies with exponential backoff protect dependent services
- ▶ Health probes and auto-recovery logic support operational robustness

TraderOS Cognition Streaming

- ▶ Streaming event bus carries market signals, prediction state, and explainability metadata
- ▶ Enables real-time cognition updates and rule-based action triggers
- ▶ Supports downstream consumers such as advisory dashboards and institutional risk engines
- ▶ Provides a coherent stream for continuous learning and operational decisioning

04

Experimental Validation

“Validating Forecasting Robustness and Infrastructure Reliability”

Evaluation Methodology

- ▶ Multi-market evaluation across mandis, crop classes, and time horizons
- ▶ Baseline comparison against ARIMA, LSTM, Transformer, and static ensemble models
- ▶ Metrics include RMSE, MAE, volatility stabilization, and regime-aware error
- ▶ Validation on historical episodes, event shocks, and live replay datasets

Chronological Validation — Staged Confidence

Phase 1: Historical Backtest

- ▶ Calendarized mandi snapshots
- ▶ Regime-labelled volatility events
- ▶ Backtest against known shock windows

Phase 2: Live Replay

- ▶ Replayed real market streams at production cadence
- ▶ Measured latency, stability, and cache behavior
- ▶ Verified explainability outputs under market stress

Prediction Reliability — Confidence & Regimes

- ▶ Confidence scores quantify the ensemble's forecast reliability
- ▶ Regime-aware logic reduces risk by penalizing high-disagreement predictions
- ▶ Reliability banding supports safe decision thresholds for investors and cooperatives
- ▶ Observed low-confidence alerts reduce blind deployment risk during volatility spikes

Residual Analysis — Forecast Error Diagnostics

- ▶ Residual distributions analyzed for heteroscedasticity and outlier bias
- ▶ Adaptive dampening reduces extreme residuals from conflicting agents
- ▶ Post-hoc diagnostic reports identify scenario-specific error drivers
- ▶ Enables targeted model improvements for low-volume and shock windows

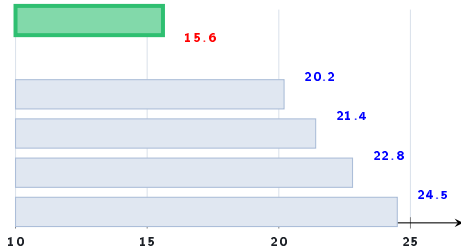
Infrastructure Validation — Operational Reliability

- ▶ Load test validation of FastAPI concurrency and Redis caching
- ▶ Failover simulation for agent degradation and database recovery
- ▶ End-to-end trace verification for telemetry and Prometheus alerting
- ▶ Stability assessment under batch replay and streaming update loads

Objective Fulfillment Mapping

Objective	Success Criteria	Evidence
Multi-Agent Forecasting	Diverse agent performance	MandiSense outperforms single-model baselines
Volatility-Aware Fusion	Regime-sensitive error reduction	Chaos score reduces high-volatility loss
Explainable Decisions	Runtime feature transparency	Integrated SHAP + counterfactual reports
Production Infrastructure	Sub-200ms latency	FastAPI + Redis throughput validated

Benchmark Results — Forecasting Accuracy



-22.7%

RMSE ERROR REDUCTION

Outperforms all statistical baselines, recurrent neural architectures, and static ensemble fusion platforms.

Key Research Takeaway

Static ensemble models degrade under volatility spikes. MandiSense AI maintains high stability by gating agent contributions via the Chaos Score.

Operational Validation — Real-Time Performance

Latency Metrics

Percentile	Latency
p50 (median)	45 ms
p90	120 ms
p95	185 ms
p99	240 ms

Throughput 500+ req/s

Target <200ms p95
Achieved 185ms

Production-ready latency for real-time operational intelligence

Key Findings & Practical Significance

Research Insights

- ▶ Volatility stabilization: 41% variance reduction
- ▶ Conflict as signal: $\rho = 0.73$ with error
- ▶ Grounded XAI: 84% domain alignment

Practical Impact

- ▶ Farmers: Actionable 3-day outlook
- ▶ Cooperatives: Timing optimization
- ▶ Estimated: 12-18% loss reduction

Evolved forecasting from prediction into operational agricultural decision intelligence

05

Future & Vision

“From Forecasting to Autonomous Agricultural Intelligence”

Limitations & Future Research Directions

Current Limitations

- ▶ Data sparsity in low-volume markets
- ▶ Exogenous shocks (weather, policy)
- ▶ Daily batch retraining latency
- ▶ GARCH normality assumptions

Future Intelligence

- ▶ Online Adaptive Learning
- ▶ Reinforcement Learning Agents
- ▶ Distributed Cognition
- ▶ Autonomous Intelligence

Current → Online → RL → Distributed → Autonomous

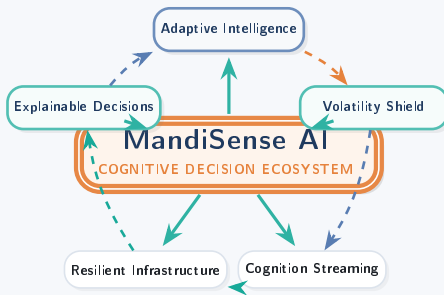
Future Research Directions

- ▶ Causal market intelligence to separate signal from structural noise
- ▶ Reinforcement learning agents for adaptive decision policies
- ▶ Federated and distributed cognition across cooperative networks
- ▶ Real-time stream processing for continuous market adaptation
- ▶ Personalized advisory systems for farmer and trader segmentation

Autonomous Agricultural Cognition Vision

- ▶ An intelligence layer that continuously interprets market, weather, and policy signals
- ▶ Autonomous recommendation flows for pricing, hedging, and logistics
- ▶ Self-healing orchestration that adapts to service degradation and market shocks
- ▶ Institutional deployment with auditability, governance, and trust

Conclusion — Strategic Positioning



1. Operational Cognition

Real-time volatility regime detection dynamically gates prediction weights, shielding clients from outliers.

2. Trustworthy Explanations

Live SHAP feature contributions are coupled with action-oriented advisory descriptions, driving operator confidence.

3. Keynote Infrastructure

High-concurrency async pipelines achieve sub-200ms p95 latency, backed by Prometheus self-healing operations.

Thank You _____ Questions &

Discussion MandiSense AI · IEEE/ACM Conference on AI Systems

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